Mattila Geometry Sets Measures Benchmark

<table>
<tr><th>Natural language statement</th><th>Domain</th><th>GroundTruth</th></tr>
<tr><td>8.19. Theorem. For a compact metric space X, H^s(X) = sup
{H^s(C) : C subset X is compact with H^s(C) < inf}.</td><td>Geometric
measure
theory</td><td>theorem mattila_8_19_compact_subsets_of_finite_hausdorff_measure
    {X : Type u} [MetricSpace X] [CompactSpace X] [MeasurableSpace X]
    [BorelSpace X]
    {s : R} (hs : 0 < s) :
    muH[s] (Set.univ : Set X) =
       C : Set X, (_ : IsCompact C), (_ : muH[s] C < inf), muH[s] C := by
  sorry</td></tr>
<tr><td>12.14. Theorem. Let A be a Borel set in Rn. (1) If dim A >
(n+1)/2, then L1(D(A)) > 0. (2) If (n-1)/2 < dim A < (n+1)/2,
then dim D(A) > dim A - (n-1)/2, where D(A) = {|x-y| : x,y in
A}.</td><td>Geometric
measure
theory</td><td>theorem mattila_12_14_falconer_distance_set
    {n : N} {A : Set (EuclideanSpace R (Fin n))} (hA : MeasurableSet A) :
    let D := {r : R | exists x in A, exists y in A, r = dist x y}
    ((((n : R>=0inf) + 1) / 2 < dimH A -> 0 < volume D) /\
      (((n : R>=0inf) - 1) / 2 < dimH A /\ dimH A < ((n : R>=0inf) + 1) / 2
      ->
        dimH A - ((n : R>=0inf) - 1) / 2 < dimH D)) := by
  sorry</td></tr>
<tr><td>6.1. Definition. The upper s-dimensional density of A at x is
Theta^{*s}(A,x) = lim sup_{r0}(2r)^{-s}H^s(A inter B(x,r)).
6.2. Theorem. Suppose A subset Rn with H^s(A) < inf. (1)
2^{-s} <= Theta^{*s}(A,x) <= 1 for H^s almost all x in A. (2)
If A is H^s measurable, Theta^{*s}(A,x) = 0 for H^s almost all
x in Rn \ A.</td><td>Geometric
measure
theory</td><td>noncomputable def upperHausdorffDensity {n : N} (s : R)
    (A : Set (EuclideanSpace R (Fin n))) (x : EuclideanSpace R (Fin n)) :
    R>=0inf :=
  limsup (fun r : R => muH[s] (A inter closedBall x r) /
    ENNReal.ofReal ((2 * r) ^ s)) (nhds[>] 0)

theorem mattila_6_2_hausdorff_density_estimates
    {n : N} {s : R} {A : Set (EuclideanSpace R (Fin n))} (hA : muH[s] A <
    inf) :
    (forall^m x dmuH[s], x in A ->
      ENNReal.ofReal (2 ^ (-s)) <= upperHausdorffDensity s A x /\
        upperHausdorffDensity s A x <= 1) /\
      (MeasurableSet A -> forall^m x dmuH[s], x notin A ->
      upperHausdorffDensity s A x = 0) := by
  sorry</td></tr>
<tr><td>15.3. Definition. A set E subset Rn is called m-rectifiable if
there exist Lipschitz maps fi : R^m -> Rn, i = 1,2,..., such
that H^m(E \ unioni fi(R^m)) = 0. A set F subset Rn is called
purely m-unrectifiable if H^m(E inter F) = 0 for every
m-rectifiable set E. 18.1. Theorem. Let A be an H^m measurable
subset of Rn with H^m(A) < inf. (1) A is m-rectifiable if and
only if H^m(P_V B) > 0 for _{n,m} almost all V in G(n,m)
whenever B is an H^m measurable subset of A with H^m(B) > 0.
(2) A is purely m-unrectifiable if and only if H^m(P_V A) = 0
for _{n,m} almost all V in G(n,m).</td><td>Geometric
measure
theory</td><td>def Grassmannian (n m : N) :=
  {V : Submodule R (EuclideanSpace R (Fin n)) // Module.finrank R V = m}

def grassmannianAction {n m : N}
    (Q : EuclideanSpace R (Fin n) equivl[R] EuclideanSpace R (Fin n))
    (V : Grassmannian n m) : Grassmannian n m :=
  <V.1.map Q.toLinearMap, (Q.finrank_map_eq V.1).trans V.2>

def IsInvariantGrassmannianMeasure {n m : N} [MeasurableSpace (Grassmannian
n m)]
    (: Measure (Grassmannian n m)) : Prop :=
  IsProbabilityMeasure /\
    forall Q : EuclideanSpace R (Fin n) equivl[R] EuclideanSpace R (Fin n),
      (forall x, ||Q x|| = ||x||) -> AEMeasurable (grassmannianAction (m :=
      m) Q) /\
        Measure.map (grassmannianAction (m := m) Q) =

def RectifiableSet (n m : N) (E : Set (EuclideanSpace R (Fin n))) : Prop :=
  exists f : N -> EuclideanSpace R (Fin m) -> EuclideanSpace R (Fin n),
    (forall j, exists K : NNReal, LipschitzWith K (f j)) /\
      muH[(m : R)] (E \ union j, range (f j)) = 0

def PurelyUnrectifiableSet (n m : N) (A : Set (EuclideanSpace R (Fin n))) :
Prop :=
  forall E : Set (EuclideanSpace R (Fin n)), RectifiableSet n m E ->
    muH[(m : R)] (A inter E) = 0

theorem mattila_18_1_besicovitch_federer_projection
    {n m : N} (hm : 0 < m) (hmn : m < n) [MeasurableSpace (Grassmannian n
    m)]
    (: Measure (Grassmannian n m)) (h : IsInvariantGrassmannianMeasure)
    {A : Set (EuclideanSpace R (Fin n))} (hA : MeasurableSet A) (hAfin :
    muH[(m : R)] A < inf) :
    (RectifiableSet n m A <-> forall B : Set (EuclideanSpace R (Fin n)),
      MeasurableSet B -> B subset A -> 0 < muH[(m : R)] B ->
        forall^m V d, 0 < muH[(m : R)] ((fun x =>
        V.1.orthogonalProjectionOnto x) '' B)) /\
      (PurelyUnrectifiableSet n m A <-></td></tr>
</table>

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<tr><td></td><td></td><td>forall^m V d, muH[(m : R)] ((fun x => V.1.orthogonalProjectionOnto
x) ’’ A) = 0) := by
sorry</td></tr>
<tr><td>7.7. Theorem. Let A subset Rn and let f: A -> R^m be a Lipschitz map. If m < s < n, then integral^{*} H^{s-m}(A inter f^{-1}{y}) dL^m y <= c(n,m) Lip(f)^m H^s(A).</td><td>Geometric measure theory</td><td>theorem mattila_7_7_lipschitz_level_sets
{n m : N} :
exists c : R>=0inf, c < inf /\ forall (s : R) (A : Set (EuclideanSpace
R (Fin n)))
(f : EuclideanSpace R (Fin n) -> EuclideanSpace R (Fin m)) (K :
NNReal),
(m : R) < s /\ s < n -> LipschitzOnWith K f A ->
(integral^- y, muH[s - m] (A inter f ^-1’ {y}) dvolume) <=
c * (K : R>=0inf) ^ (m : R) * muH[s] A := by
sorry</td></tr>
<tr><td>Definition. The s-dimensional Hausdorff content of A is H^s_inf(A) = inf sumi d(Ei)^s, where the infimum is over all countable covers A subset unioniEi. 8.8. Theorem. Let B be a Borel set in Rn. Then H^s(B) > 0 if and only if there exists mu in M(B) such that mu(B(x,r)) < r^s for x in Rn and r > 0. Moreover, we can find mu so that mu(B) > cH^s_inf(B), where c > 0 depends only on n.</td><td>Geometric measure theory</td><td>noncomputable def hausdorffContent {X : Type u} [PseudoMetricSpace X]
(s : R) (A : Set X) : R>=0inf :=
U : N -> Set X, (_ : A subset union i, U i),
sum’ i, ENNReal.ofReal (diam (U i)) ^ s

theorem mattila_8_8_frostman_lemma
{n : N} :
exists c : R>=0inf, 0 < c /\ c < inf /\
forall (s : R) (B : Set (EuclideanSpace R (Fin n))),
0 < s -> MeasurableSet B ->
(0 < muH[s] B <-> exists mu : Measure (EuclideanSpace R (Fin n)),
IsFiniteMeasureOnCompacts mu /\ Measure.InnerRegular mu /\ mu
!= 0 /\
mu B^c = 0 /\
forall x : EuclideanSpace R (Fin n), forall r : R, 0 < r ->
mu (closedBall x r) < ENNReal.ofReal (r ^ s)) /\
exists mu : Measure (EuclideanSpace R (Fin n)),
IsFiniteMeasureOnCompacts mu /\ Measure.InnerRegular mu /\ mu
!= 0 /\
mu B^c = 0 /\
(forall x : EuclideanSpace R (Fin n), forall r : R, 0 < r ->
mu (closedBall x r) < ENNReal.ofReal (r ^ s)) /\
c * hausdorffContent s B <= mu B := by
sorry</td></tr>
<tr><td>Definition. The Riesz s-energy of mu is I_s(mu) = integralintegral|x-y|^{-s} dmux dmuy. The Grassmannian G(n,m) is the space of m-dimensional linear subspaces of Rn, and _{n,m} is its orthogonally invariant probability measure. 9.7. Theorem. Let mu be a Radon measure on Rn with compact support and with I_m(mu) < inf. Then P_{V#}mu << H^m for _{n,m} almost all V in G(n,m) and integral_V D(P_{V#}mu,u)^2 dH^m u d_{n,m}V < cI_m(mu), where c is a constant depending only on n and m.</td><td>Geometric measure theory</td><td>noncomputable def rieszEnergy {n : N} (s : R)
(mu : Measure (EuclideanSpace R (Fin n))) : R>=0inf :=
integral^- x, (integral^- y, ENNReal.ofReal (dist x y)^-1 ^ s dmu) dmu

def Grassmannian (n m : N) :=
{V : Submodule R (EuclideanSpace R (Fin n)) // Module.finrank R V = m}

def grassmannianAction {n m : N}
(Q : EuclideanSpace R (Fin n) equivl[R] EuclideanSpace R (Fin n))
(V : Grassmannian n m) : Grassmannian n m :=
<V.1.map Q.toLinearMap, (Q.finrank_map_eq V.1).trans V.2>

def IsInvariantGrassmannianMeasure {n m : N} [MeasurableSpace (Grassmannian
n m)]
(: Measure (Grassmannian n m)) : Prop :=
IsProbabilityMeasure /\
forall Q : EuclideanSpace R (Fin n) equivl[R] EuclideanSpace R (Fin n),
(forall x, ||Q x|| = ||x||) -> AEMeasurable (grassmannianAction (m :=
m) Q) /\
Measure.map (grassmannianAction (m := m) Q) =

theorem mattila_9_7_projection_energy
{n m : N} [MeasurableSpace (Grassmannian n m)]
{mu : Measure (EuclideanSpace R (Fin n))} (: Measure (Grassmannian n
m))
(h : IsInvariantGrassmannianMeasure)
(hmu : IsFiniteMeasureOnCompacts mu /\ Measure.InnerRegular mu /\
IsCompact mu.support /\
rieszEnergy (m : R) mu < inf) :
(forall^m V d,
Measure.map (fun x => V.1.orthogonalProjectionOnto x) mu << muH[(m</td></tr>
</table>

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<tr><td></td><td></td><td>: R)]) /\
exists (density : forall V : Grassmannian n m, V.1 -> R>=0inf) (c :
R>=0inf), c < inf /\
(forall^m V d, Measure.map (fun x => V.1.orthogonalProjectionOnto
x) mu =
muH[(m : R)].withDensity (density V)) /\
integral^- V, integral^- x, density V x ^ (2 : R) dmuH[(m : R)] d <=
c * rieszEnergy (m : R) mu := by
sorry</td></tr>
<tr><td>Definition. The Grassmannian G(n,k) is the space of
k-dimensional linear subspaces of Rn, and _{n,k} is its
orthogonally invariant probability measure. 10.10. Theorem.
Let m < t < n and let A subset Rn be a Borel set with 0 <
H^t(A) < inf. Then for all W in G(n,n-m), H^{t-m}(AinterW_a) <
inf for H^m almost all a in W^bot, and for _{n,n-m} almost all
W in G(n,n-m), H^m({a in W^bot : dim(AinterW_a)=t-m}) > 0.</td><td>Geometric
measure
theory</td><td>def Grassmannian (n m : N) :=
{V : Submodule R (EuclideanSpace R (Fin n)) // Module.finrank R V = m}

def grassmannianAction {n m : N}
(Q : EuclideanSpace R (Fin n) equivl[R] EuclideanSpace R (Fin n))
(V : Grassmannian n m) : Grassmannian n m :=
<V.1.map Q.toLinearMap, (Q.finrank_map_eq V.1).trans V.2>

def IsInvariantGrassmannianMeasure {n m : N} [MeasurableSpace (Grassmannian
n m)]
(: Measure (Grassmannian n m)) : Prop :=
IsProbabilityMeasure /\
forall Q : EuclideanSpace R (Fin n) equivl[R] EuclideanSpace R (Fin n),
(forall x, ||Q x|| = ||x||) -> AEMeasurable (grassmannianAction (m :=
m) Q) /\
Measure.map (grassmannianAction (m := m) Q) =

theorem mattila_10_10_plane_sections
{n m : N} {t : R} {A : Set (EuclideanSpace R (Fin n))}
[MeasurableSpace (Grassmannian n (n - m))]
(: Measure (Grassmannian n (n - m)))
(h : IsInvariantGrassmannianMeasure)
(hmt : (m : R) < t /\ t < n) (hA : MeasurableSet A)
(hAfinite : muH[t] A < inf) (hApos : 0 < muH[t] A) :
let slice := fun (W : Grassmannian n (n - m)) (a : W.1) =>
A inter {x | x - (a : EuclideanSpace R (Fin n)) in W.1}
(forall W : Grassmannian n (n - m),
forall^m a d(muH[(m : R)] : Measure W.1),
(muH[t - m] : Measure (EuclideanSpace R (Fin n))) (slice W a) <
inf) /\
forall^m W : Grassmannian n (n - m) d,
0 < muH[(m : R)] {a : W.1 |
dimH (slice W a) = ENNReal.ofReal (t - m)} := by
sorry</td></tr>
<tr><td>14.10. Theorem. Let s be a positive number. Suppose that there
exists a Radon measure mu on Rn such that the density
Theta^s(mu,a) exists and is positive and finite in a set of
positive mu measure. Then s is an integer.</td><td>Geometric
measure
theory</td><td>theorem mattila_14_10_marstrand_density_integer
{n : N} {s : R} {mu : Measure (EuclideanSpace R (Fin n))}
(hs : 0 < s) (hmu : IsFiniteMeasureOnCompacts mu /\
Measure.InnerRegular mu)
(hdensity : exists E : Set (EuclideanSpace R (Fin n)), 0 < mu E /\
forall x in E,
exists theta : R>=0inf, 0 < theta /\ theta < inf /\
Tendsto (fun r : R => mu (closedBall x r) / ENNReal.ofReal ((2 * r)
^ s))
(nhds[>] 0) (nhds theta)) :
exists m : N, s = m := by
sorry</td></tr>
<tr><td>15.3. Definition. A set E subset Rn is called m-rectifiable if
there exist Lipschitz maps fi : R^m -> Rn, i = 1,2,..., such
that H^m(E \ unionifi(R^m)) = 0. 15.7. Definition. We say that
a subset E of Rn is m-linearly approximable if for H^m almost
all a in E the following holds: if eta is a positive number,
there are positive numbers r0 and lambda and an affine m-plane
W such that a in W and for any 0 < r < r0, H^m(E inter
B(x,etar)) >= lambdar^m for x in W inter B(a,r), and H^m(E
inter B(a,r) \ W(etar)) < etar^m. 15.17. Definition. Let A
subset Rn, a in Rn and V in G(n,m). We say that V is an
approximate tangent m-plane for A at a if Theta^{*m}(A,a) > 0
and for all 0 < s < 1, r^{-m}H^m(A inter B(a,r) \ X(a,V,s)) ->
0 as r 0. 15.19. Theorem. Let E be an H^m measurable subset</td><td>Geometric
measure
theory</td><td>def Grassmannian (n m : N) :=
{V : Submodule R (EuclideanSpace R (Fin n)) // Module.finrank R V = m}

def RectifiableSet (n m : N) (E : Set (EuclideanSpace R (Fin n))) : Prop :=
exists f : N -> EuclideanSpace R (Fin m) -> EuclideanSpace R (Fin n),
(forall j, exists K : NNReal, LipschitzWith K (f j)) /\
muH[(m : R)] (E \ union j, range (f j)) = 0

def LinearlyApproximableSet (n m : N) (E : Set (EuclideanSpace R (Fin n)))
: Prop :=
forall^m a dmuH[(m : R)].restrict E, forall eta : R, 0 < eta ->
exists V : Grassmannian n m, exists r0 c : R, 0 < r0 /\ 0 < c /\
forall r : R, 0 < r -> r < r0 -></td></tr>
</table>

Natural language statement	Domain	GroundTruth

of Rn with H^m(E) < inf. Then the following are equivalent: (1) E is m-rectifiable. (2) E is m-linearly approximable. (3) For H^m almost all a in E there is a unique approximate tangent m-plane for E at a. (4) For H^m almost all a in E there is some approximate tangent m-plane for E at a.		<pre> (forall x, x - a in V.1 -> x in closedBall a r ->
		          ENNReal.ofReal (c * r ^ m) <= muH[(m : R)] (E inter closedBall x
		        (eta * r))) /\
		        muH[(m : R)] (E inter closedBall a r inter
		          {x | eta * r <= infDist (x - a) (V.1 : Set (EuclideanSpace R (Fin
		        n)))}) <
		            ENNReal.ofReal (eta * r ^ m)
		
		def IsApproximateTangentPlane {n m : N} (E : Set (EuclideanSpace R (Fin n)))
		    (a : EuclideanSpace R (Fin n)) (V : Grassmannian n m) : Prop :=
		  0 < upperHausdorffDensity (m : R) E a /\
		    forall s : R, 0 < s -> s < 1 ->
		      Tendsto (fun r : R =>
		        muH[(m : R)] (E inter closedBall a r inter
		          {x | s * dist x a <= infDist (x - a)
		            (V.1 : Set (EuclideanSpace R (Fin n)))}) /
		            ENNReal.ofReal (r ^ (m : R))) (nhds[>] 0) (nhds 0)
		
		theorem mattila_15_19_rectifiability_tangent_planes
		    {n m : N} {E : Set (EuclideanSpace R (Fin n))}
		    (hEmeas : MeasurableSet E) (hEfinite : muH[(m : R)] E < inf) :
		    List.TFAE [RectifiableSet n m E, LinearlyApproximableSet n m E,
		      forall^m a dmuH[(m : R)].restrict E,
		        exists! V : Grassmannian n m, IsApproximateTangentPlane (m := m) E
		        a V,
		      forall^m a dmuH[(m : R)].restrict E,
		        exists V : Grassmannian n m, IsApproximateTangentPlane (m := m) E a
		        V] := by
		  sorry</pre>